

A Systematic Literature Review on Website Design and Its Influence on User Satisfaction: A PRISMA-Guided Global Synthesis

Jelai Fabricante
Department of Information Technology
Alangalang, Philippines
jelaifabricante@gmail.com

Abstract— This review examined how website design contributes to user satisfaction by exploring the interaction between design features and individual user characteristics at a time when digital platforms had become essential in global communication, work, and service delivery. Prior studies pointed out the importance of visual appeal, usability, and functionality; however, two issues remained insufficiently explored: the role of individual differences, particularly digital literacy, and the consistent integration of accessibility in design evaluations. Evidence across the literature showed that these factors were often neglected, limiting the accuracy and applicability of earlier findings. To understand these concerns in a broader context, the review incorporated studies from multiple regions, reflecting diverse website users and design practices. Guided by the PRISMA 2020 framework, fifty peer-reviewed studies were systematically selected and analyzed through thematic coding in alignment with three research objectives. All stages of screening and synthesis followed ethical standards for secondary research. The results showed that usability and satisfaction dominated academic attention, while accessibility, visual design, cultural differences, and user skills appeared far less frequently. These patterns indicated uneven progress in the field and underscored the need for more inclusive approaches that capture the varied experiences of global users.

Keywords: website design, user satisfaction, usability, accessibility, user experience

I. INTRODUCTION

Over the past decade, digital technologies have completely changed the function and purpose of websites. Instead of serving as pages of information, they have now become an environment that supports communication, business, and even education. As more organizations rely mainly on digital tools, website design has become an important factor that influences how users engage with and trust online platforms. In 2024, global statistics [1] indicated that over 5.35 billion individuals, or approximately 66% of the world's population, actively used the internet every day. Considering the widespread online activity, it was important to understand how the quality of

website design affects user experience and satisfaction. Prior studies have shown that users often form judgments about a website's reliability, credibility, and aesthetic value within milliseconds of first exposure [2][3]. However, researchers continued to debate which design features, such as visual appeal, usability, interactivity, or accessibility, had the strongest influence on satisfaction, especially when considering different cultures, devices, and user preferences.

Existing research frequently focused on individual design dimensions, such as navigation efficiency or color schemes, without examining their interactive effects on cognitive and emotional user responses [4], [5], [6]. These studies mentioned usability or design satisfaction, but don't include emotional or cognitive testing. Studies grounded in frameworks such as the Technology Acceptance Model (TAM) or WebQual [7], [8], [9], [10] primarily concentrated on variables such as perceived ease of use or service quality, producing results that are often context-dependent and difficult to generalize. Moreover, user demographics, digital literacy, and device habits were typically treated as secondary factors, even though they can significantly affect satisfaction levels. These limitations highlight the need for a comprehensive synthesis that integrates behavioral, psychological, and technological perspectives to explain how website design holistically shapes user experience.

To address these gaps, this systematic literature review was conducted using the PRISMA framework to consolidate and analyze findings from fifty (50) peer-reviewed studies across multiple disciplines. The review pursued three primary objectives: first, to examine how individual differences such as user demographics, digital literacy, and personal preferences had influenced perceptions of website design and user satisfaction; second, to evaluate the balance between visual appeal, functional usability, and accessibility in shaping overall user experience; and third, to synthesize global research trends to identify convergent patterns, emerging themes, and persisting gaps in the literature. By incorporating studies from

diverse geographic and cultural contexts, the review provided a comprehensive synthesis of how design characteristics collectively shaped satisfaction outcomes. Through these objectives, the study developed an integrated understanding of the multidimensional relationship between website design and user satisfaction, offering a structured foundation for advancing evidence-based and user-centered web design research.

The significance of this review extends beyond academic inquiry to encompass global sustainability and digital equity. By promoting user-centered, inclusive, and accessible web design practices, the study supports the United Nations Sustainable Development Goals (SDGs), particularly SDG 9 (Industry, Innovation, and Infrastructure) and SDG 12 (Responsible Consumption and Production). Sustainable digital infrastructures rely on design frameworks that strike a balance between aesthetic appeal, efficiency, and accessibility, ensuring that technological advancement benefits all user groups equitably. The synthesized findings contribute to a deeper understanding of how evidence-based design can enhance user satisfaction, strengthen trust, and promote responsible innovation in digital ecosystems. Ultimately, this review provides both theoretical and empirical foundations for developing sustainable web design models that advance user experience excellence and promote long-term technological resilience in an increasingly interconnected global society.

II. REVIEW OF RELATED LITERATURE

A. *Influence of Individual Differences on Perception and Satisfaction*

Prior studies have consistently emphasized that user satisfaction is not uniform across all audiences, as individual differences such as age, gender, digital literacy, and cultural background significantly shape website experience. Guo et al. [11] and Lun et al. [7] found that demographic factors influenced the way users perceived interface clarity and navigation ease, particularly when comparing beginner and expert internet users. Ingriana [12] further demonstrated that cultural norms affect how people interpret aesthetic balance and information density in websites, leading to varied satisfaction outcomes between Western and Asian users. Malik et al. [6] revealed that users with higher digital literacy often show less tolerance for slow-loading pages or broken links, directly linking usability perceptions to satisfaction levels. These studies indicated that website design evaluations were subjective and deeply influenced by users' cognitive and social backgrounds.

Additional evidence from Liu and Park [13] identified personalization and adaptive design as emerging strategies to bridge demographic differences, reinforcing that user satisfaction could be improved through adaptive user experience approaches. These findings confirmed that user-centered design required an understanding of individual variability rather than a one-size-fits-all layout.

B. *Balance Between Visual Appeal, Functional Usability, and Accessibility*

A substantial body of research examined how visual aesthetics, usability, and accessibility interact to determine website satisfaction. Santoso [14] and Wijaya et al. [8] confirmed that color harmony, font readability, and page symmetry influenced users' first impressions and trust formation. Moreno et al. [15] and Lun et al. [7] added that a visually engaging interface alone was insufficient to ensure satisfaction if it compromised loading time or functional clarity. These findings strengthened the multidimensional nature of design evaluation, where aesthetic pleasure and instrumental performance needed to coexist.

Studies on accessibility expanded this perspective. Goyal & Singh [16] found that mobile-friendly designs and features supporting users with disabilities improved perceived inclusiveness and satisfaction. Zuo et al. [17] demonstrated that simplified layouts enhanced task completion rates among older adults, thereby improving satisfaction. In contrast, overly complex visuals or interactive clutter reduced cognitive efficiency and discouraged engagement [18]. These works highlight that sustainable satisfaction depends on achieving a balance between beauty, usability, and accessibility.

Yin et al. [19] provided further evidence that objective performance metrics such as click path efficiency and error rate are closely linked to perceived satisfaction. While studies of Horosh [20] and Chiu & Yang [21] discovered that responsiveness and consistency of navigation menus were among the most critical usability indicators across different platforms. These converging findings support that satisfaction was maximized when design elements complemented cognitive fluency and emotional resonance simultaneously.

C. *Global Trends, Emerging Innovations, and Persistent Gaps*

Recent investigations explored website design innovation from global and cross-cultural perspectives. Wang and Inkuer [22] examined comparative satisfaction models across Asia, Europe, and North America, finding consistent importance of responsiveness and content relevance, though aesthetic preferences varied regionally. Sajjan et al. [23] and Zaheer [24] analyzed emerging innovations such as AI-driven personalization, adaptive layouts, and emotion-aware interfaces that adjusted tone and visuals to the user's mood. These advancements reflected a global shift towards data-driven personalization to enhance user satisfaction.

However, despite these advancements, several gaps remained. Santoso [14] and Dobbala & Lingolu [25] argued that while aesthetic and functional features often dominated design evaluations, the social and ethical aspects of inclusivity, such as equitable access and cultural adaptability, remained underrepresented. Similarly, Gupta et al. [26] highlighted that although accessibility compliance had improved, the

integration of inclusive design principles across diverse user populations was inconsistent. Lewis et. al [27] further emphasized that satisfaction was not static but evolved through repeated interactions, underlining the absence of longitudinal studies examining how sustained exposure to interface design influenced user trust and long-term engagement. These persisting gaps indicated the need for research frameworks that incorporate temporal user experience, inclusivity, and accessibility as core components of satisfaction evaluation.

Additionally, global research called attention to many existing models, such as TAM, WebQual, and UX heuristics were often applied in isolation [7], leading to fragmented results. Recent integrative frameworks proposed by Liu and Park [13] advocated for multidimensional analysis that captured emotional, behavioral, and cognitive components simultaneously. These findings collectively emphasized the need for consolidated, evidence-based models to unify theoretical fragmentation in web design satisfaction research.

Across the reviewed fifty (50) studies, consistent patterns emerged linking website design to user satisfaction through the mediating influence of user experience. Prior research validated that user perceptions were strongly affected by individual differences in culture, literacy, and motivation. Meanwhile, functional usability and visual appeal worked collaboratively to produce emotional engagement and efficiency. Despite increasing methodological sophistication, gaps remained in integrating accessibility, personalization, and longitudinal satisfaction tracking. The literature confirmed that an evidence-based, user-centered approach was essential to improving satisfaction outcomes. These findings provided the theoretical and empirical foundation for the present systematic review, which sought to unify fragmented design dimensions into a cohesive model explaining how website design influences satisfaction through user experience.

III. METHODOLOGY

The conceptual framework derived from this systematic literature review illustrates the multidimensional relationship among website design characteristics, individual user differences, and user satisfaction outcomes. The synthesis of prior studies revealed that design factors such as visual aesthetics, usability, functionality, and accessibility directly influence satisfaction, while individual differences, including demographics, digital literacy, and cultural background, moderate these relationships.

In this framework, website design characteristics function as the independent variables, encompassing layout, content organization, navigational structure, and accessibility compliance. Individual differences serve as moderating factors, shaping how users perceive and evaluate design quality. Users with higher digital literacy tend to emphasize functionality and performance, whereas those from collectivist cultures may value harmony and aesthetic balance. The user satisfaction

outcomes represent the dependent variables, comprising both cognitive (e.g., efficiency, ease of use) and affective (e.g., trust, enjoyment) responses. Adaptive and personalized design mechanisms further mediate the relationship between design and satisfaction, enabling interfaces to adjust to user behavior and contextual conditions.

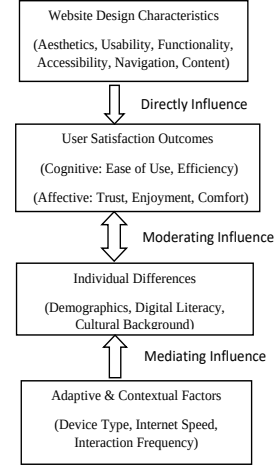


Figure 1. Conceptual Framework of the Study

Figure 1 illustrates the interaction among website design characteristics, individual user differences, and satisfaction outcomes. Website design features directly affect user satisfaction, while demographic and cultural factors moderate user perceptions of these designs. Adaptive and contextual variables such as device type and internet speed mediate the relationship, highlighting satisfaction as a dynamic process shaped by both design quality and user diversity.

Moreover, this review applied a Systematic Literature Review (SLR) approach guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) framework. This method was selected for its structured and transparent process in identifying, screening, evaluating, and synthesizing published literature relevant to the relationship between website design and user satisfaction. The PRISMA protocol allowed for a consistent evaluation of sources, ensuring that all stages of the review adhered to academic rigor and replicability [28]. To provide clarity on the review process, the following subsections describe the data sources used, the search strategy applied, and the criteria followed during screening.

A. Data Source and Search Strategy

Data collection was conducted using the Publish or Perish (PoP) software, which extracted bibliographic information and citation metrics from Google Scholar and other scholarly databases. The search employed a combination of Boolean operators and targeted keywords, including “website design”, “user experience”, “user satisfaction”, “usability”, and “UX design”, to ensure comprehensive coverage of relevant studies. The inclusion was limited to peer-reviewed articles published

between 2020 and 2025, written in English, and available in full text.

To maintain focus and reliability, the search was confined to studies examining how elements of website design, such as layout, color scheme, typography, navigation, and functionality, affect user satisfaction in various contexts, including education, business, and digital platforms.

B. Inclusion and Exclusion Criteria

Following the PRISMA guidelines [28], the articles underwent several phases of evaluation: identification, screening, eligibility, and inclusion. A total of two hundred fourteen (214) records were initially identified from electronic databases, with twenty-one (21) additional papers located through manual searches. After the removal of duplicates ($n = 1$) and irrelevant records ($n = 186$), forty-nine (49) articles were subjected to a full-text screening. From these, thirty-nine (39) studies met the inclusion criteria and were incorporated into the final synthesis.

Exclusion was applied to papers that (1) represented additional review studies already assessed, (2) demonstrated substantial methodological or contextual limitations, or (3) lacked sufficient alignment with the research objectives. These steps are summarized in the PRISMA Flowchart (Figure 2), which visually depicts the systematic process of article selection and elimination.

C. Data Extraction and Analysis

The selected studies were coded and analyzed to identify key themes, methods, and findings. Each study was examined for its research design, target population, analytical framework, and major results. A qualitative thematic analysis was applied to synthesize patterns across studies, allowing the identification of converging and diverging insights concerning user satisfaction determinants.

Data were organized into three thematic lenses aligned with the research objectives: (1) Individual differences (e.g., digital literacy, demographics, and personal preferences); (2) Design balance among visual appeal, functionality, and accessibility; and (3) Global trends reflecting cross-cultural perspectives on website usability and satisfaction. This integrative approach enabled a comprehensive understanding of how diverse design characteristics collectively influence satisfaction outcomes.

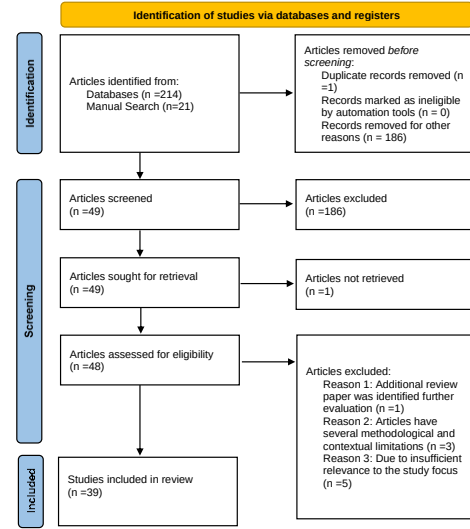


Figure 2. PRISMA 2020 Flow Diagram of the Systematic Review Process

Figure 2 presents a visual representation that enhances methodological transparency and provides a concise overview of the inclusion and exclusion procedure for the reviewed studies.

D. Ethical Standard

This review was conducted in accordance with ethical principles governing academic research. All secondary sources were appropriately acknowledged and cited. Since the study did not involve human or animal participants, it posed no ethical risks. To maintain research integrity, Grammarly was used for grammatical accuracy and clarity, while similarity index tools (e.g., Turnitin or Quetext) verified that the manuscript has maintained AI-generated-free and plagiarism-free standards. In addition, Artificial intelligence tools, including ChatGPT, were used solely to assist in language refinement and formatting, but not in data interpretation or analytical reasoning, which ensures the originality and academic authenticity of the final output.

IV. RESULT

This section presents the findings of the systematic literature review according to the three research objectives. Each objective was organized into its own subsection and was supported by visual summaries that supported the synthesized findings.

1. Individual Preferences

The researcher examined how individual differences such as demographics, digital literacy, and personal preferences had influenced perceptions of website design and user satisfaction. To determine the extent to which these dimensions appeared in the reviewed studies, the frequency of topic coverage was organized and presented in Table 1.

Table 1. Gap Analysis Summary

This table shows the number of studies that mentioned each objective-related theme, allowing for the identification of areas with strong or limited representation.

Topic	Studies Mentioning
Satisfaction	34
Usability or Functionality	30
Demographics	26
Visual Appeal or Aesthetics	13
Accessibility	10
Preferences	5
Digital literacy	0

The distribution showed that satisfaction (34) and usability or functionality (30) were the most commonly addressed themes. Demographics appeared in 26 studies, while visual appeal or aesthetics appeared in only 13. Accessibility was noted in 10 studies, and user preferences in only 5. However, digital literacy did not appear in any of the reviewed studies, revealing a substantial gap in the literature. This absence indicated that many researchers assumed a uniform level of user competence, overlooking realistic differences in skill and experience.

2. Visual Appeal, Usability, and Accessibility

To address the second research objective, the researcher evaluated how visual appeal, usability, and accessibility shaped the user experience. This included identifying recurring themes and assessing the methodological approaches employed in the studies, as shown in Figures 3 and 4.

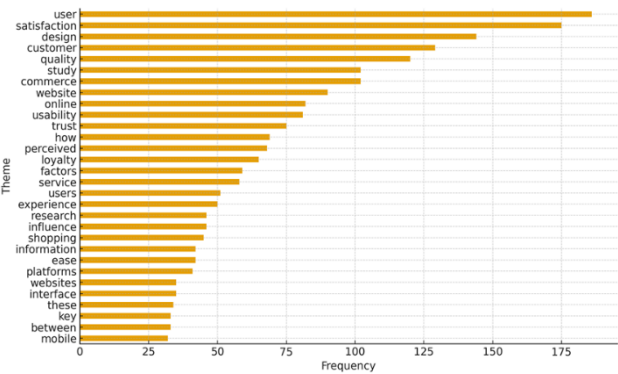


Figure 3. Theme Frequency

Figure 3 shows that usability and satisfaction were the most frequently emphasized themes across titles, abstracts, and

findings, whereas accessibility-related terms appeared considerably less often.

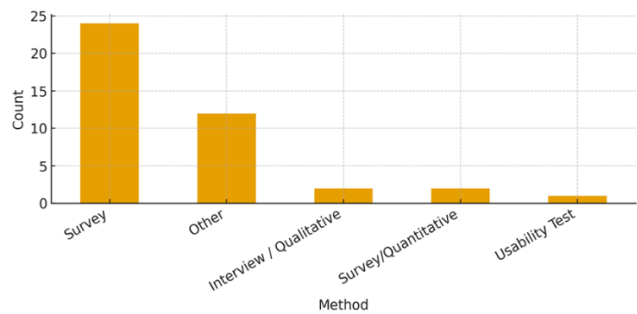


Figure 4. Method Distribution

Figure 4 illustrates the range of methodologies used in the included studies. Surveys and usability testing were the dominant research methods, reflecting a focus on performance measures rather than inclusive or experiential aspects.

3. Research Trends, Convergent Patterns, and Gaps

The researcher then synthesized broader research trends to identify convergent patterns, emerging themes, and persisting gaps. This analysis drew on publication years and keyword relationships.

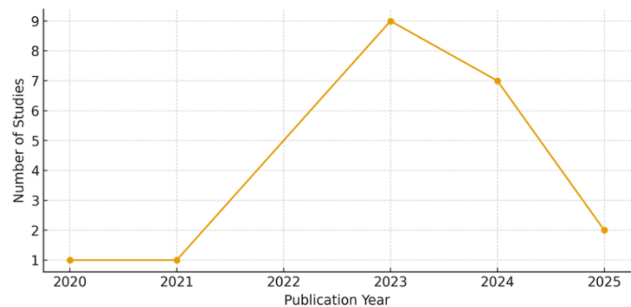


Figure 5. Publication Trends

Figure 5 presents variations in annual publication output, showing uneven research interest.

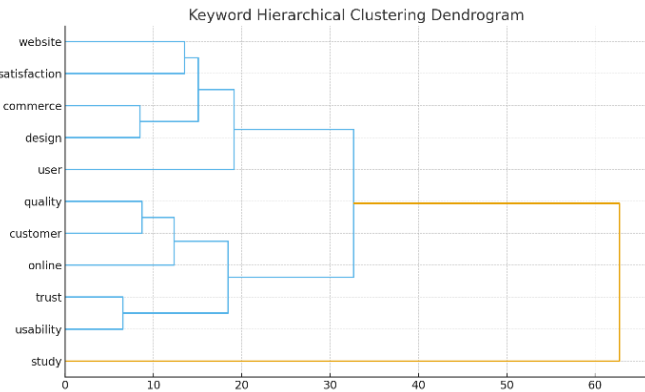


Figure 6. Keyword Hierarchical Clustering Dendrogram

Figure 6 displays group keywords based on how often they appeared together within the studies. The largest merge, shown in orange, reflects the most distant connection from the other groups, meaning this theme had a broader but less specific link to the rest. Meanwhile, the closer clusters centered around usability, trust, online experience, customer-related factors, and satisfaction, topics that appeared more consistently across the literature.

V. DISCUSSION

The purpose of this review was to synthesize existing research on how individual factors, design elements, and broader scholarly trends had shaped user experience in website environments. The consolidated findings revealed recurring patterns and persistent gaps that limited the field's ability to represent diverse user populations and emerging design concerns. Interpreting these results in relation to the research objectives provided a more comprehensive view of the dominant directions of the field, strengths, and weaknesses of current website design, and clarified areas where current academic study remained underexplored.

The researcher first examined how individual preferences and user characteristics had influenced website experience. The gap analysis demonstrated a disproportionate emphasis on satisfaction and usability, while demographics, personal preferences, and accessibility received substantially less attention. Most notably, none of the reviewed studies addressed digital literacy. The complete absence of digital literacy across the reviewed studies was particularly significant. This omission suggested that many researchers approached user groups as though they possessed uniform levels of digital skill, which may have led to overly generalized conclusions about satisfaction and engagement. Because technological competence affects navigation strategies, error tolerance, and perceptions of usability, excluding digital literacy limited the explanatory power of existing models. Similarly, the limited discussion of personal preferences indicated that many studies tended to conceptualize users in a standardized way, rather than acknowledging the variability in motivations and expectations that shape real-world website interactions. These findings reflected an important conceptual gap: the literature has not yet fully captured the diversity of user behaviors that influence online experience.

Beyond individual differences, the analysis of design-related factors revealed a similar imbalance. Usability and satisfaction remained the most emphasized themes, indicating a long-standing focus on efficiency, clarity, and task performance. In contrast, accessibility, an essential component in equitable digital design, appeared inconsistently across studies. The methodological distribution supported this trend as surveys and usability tests dominated the research, strengthening an emphasis on performance metrics and user self-reports. Although these methods offer valuable insights into performance and perceived ease of use, they provide only a partial understanding of user experience by overlooking aesthetic perceptions, emotional responses, and disability-related needs. As a result, the existing body of research offers a strong foundation for understanding functional dimensions of website design but presents a more limited account of the

holistic and inclusive elements that contemporary UX standards now prioritize. This imbalance suggests that current frameworks do not fully reflect the complexity of how users engage with digital interfaces.

The synthesis of research trends provided additional insight into the field's development. Variations in publication outputs suggested that academic interest in website experience tended to respond to shifts in digital technology rather than follow a steady direction of growth. Despite this variability, the keyword co-occurrence analysis showed strong thematic clustering around usability, trust, satisfaction, customer factors, and online experience, revealing a cohesive core around which much of the literature had been organized. These clusters reflected the persistence of a user-centered but performance-driven framework for evaluating website quality. In contrast, the most distant merge in the dendrogram indicated that certain concepts remained broad, insufficiently integrated, or lacking in conceptual clarity. This imbalance suggested that although some aspects of website experience had been thoroughly theorized, others remained underdeveloped or were treated only as secondary concerns.

The findings portrayed a field that has made considerable progress in defining and evaluating usability and satisfaction but has not yet achieved comparable depth in its treatment of individual differences, aesthetic perceptions, or accessibility considerations. The consistent absence of digital literacy, the limited discussion of user preferences, and the inconsistent integration of inclusive design highlight the need for more diverse and methodologically varied approaches. Future research would benefit from examining a wider range of user characteristics, adopting mixed and innovative methods capable of capturing both performance-based and experiential dimensions, and positioning accessibility as a foundational rather than peripheral element of website evaluation. Addressing these gaps would contribute to a more comprehensive and equitable body of knowledge, better aligning website design research with the realities of an increasingly diverse and digitally dependent global population.

VI. CONCLUSION

This systematic literature review gathered findings from fifty (50) peer-reviewed studies to understand how website design features and individual user differences had influenced user satisfaction in different digital settings. The review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework, which helped the researcher select, screen, and analyze the studies in a consistent and organized way. This approach made it possible to identify key patterns in the literature and recognize areas that required further attention.

The findings showed that individual differences influenced how users interacted with websites, but these factors were not discussed evenly across the studies. Demographic characteristics appeared in several articles, while personal preferences and digital literacy were mentioned far less often. The complete absence of digital literacy across the reviewed studies was especially important because it suggested that many

studies assumed that all users shared similar levels of skill when using digital tools. As a result, many published conclusions about user satisfaction may not have represented the range of abilities and backgrounds found in actual digital audiences.

As the analysis progressed to design factors, the researcher found that usability remained the most emphasized aspect of website evaluation. Many studies focused on efficiency, clarity, and ease of use. Visual appeal received less attention, and accessibility was mentioned only sometimes. This pattern showed that most studies placed greater importance on how well a website functioned rather than on how visually engaging or inclusive it was. Although usability remains central to user experience, the limited attention given to accessibility and visual elements suggests that the literature offered an incomplete description of how users assess and engage with websites.

The review also emphasized how research trends changed over time. Publication counts rose and fell, suggesting that interest in website experience shifted alongside technological developments. Common keywords such as usability, trust, satisfaction, and online experience continued to guide much of the research, while other ideas appeared only occasionally and had not yet developed into consistent topics of study.

The combined evidence showed that the field had made solid progress in understanding usability and satisfaction. However, areas such as individual differences, cultural backgrounds, and accessibility needs still require more attention, even though they clearly influence how users interact with websites. These observations suggested that future studies would benefit from using a wider range of methods, involving more diverse groups of users, and placing stronger emphasis on inclusive design. Doing so would support a more accurate and fair understanding of how people with different abilities and experiences engage with digital platforms.

By bringing together insights from many studies, this review provided a clearer understanding of how website design and user diversity worked together to influence satisfaction. The findings pointed out that website design should be approached not only as a technical activity but also as one that considers the needs, skills, and expectations of different users. This perspective can guide future work toward creating website designs that better reflect the experiences and requirements of a broad and growing digital population.

VII. RECOMMENDATION

Future studies should prioritize a more comprehensive understanding of user diversity by integrating variables that have received limited attention, such as digital literacy, accessibility needs, and culturally shaped design preferences. As digital platforms continue to advance, it becomes increasingly important to examine how varying levels of digital competence influence navigation behavior, trust, and user satisfaction. Focusing on these individual differences will allow future research to better reflect the realities of diverse digital populations and to support the development of designs that

respond to users with different skills, backgrounds, and expectations.

Researchers are also encouraged to adopt more diverse and contemporary methodological approaches rather than relying primarily on surveys and usability testing. The use of mixed methods, including eye-movement studies, affective analytics, longitudinal tracking, and adaptive user profiling, can offer deeper insights into both functional and emotional aspects of user experience. As interface technologies become more personalized and multimodal, subsequent investigations could consider exploring how visual design, accessibility features, and adaptive interface behaviors interact to shape overall satisfaction. Broadening methodological strategies will strengthen the field's capacity to capture the complexity of real-world user interactions.

In addition, future-proofing research requires stronger attention to inclusive and ethical design principles. Accessibility should be treated as a foundational requirement rather than an optional element, particularly as global reliance on digital systems continues to grow. It will also be essential for future studies to investigate how emerging technologies, such as AI-generated content, automated personalization, and cross-device ecosystems, influence user expectations and satisfaction outcomes. Aligning research priorities with technological developments and inclusive design standards will help generate insights that are both equitable and responsive to the rapidly evolving digital environment.

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